SWEET.TV TV Program Recommendation

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Repository: https://github.com/PetrVorozhtsov/RecSys-LLMs/tree/exam-sweettv/exam_sweettv

1. Data and protocol

View logs were joined to labelled schedule rows by equal channel and interval overlap. A view was accepted when the overlap was at least 60 seconds and at least 10% of the scheduled programme duration. Rows with tv_show_id = 0 were excluded; when several rows overlapped, the row with the largest overlap was selected. Training used weeks 11–20, validation used weeks 21–30, and the final catalogue used labelled programmes from weeks 11–42. Test-period user behaviour was not used for fitting.

Join audit: 5,821,987 input views; 502,213 matched labelled interactions; 1,671,100 unmatched or unlabelled; 3,648,674 views from users outside the submission template.

2. Methods

The popularity reference ranks programmes by the number of unique training viewers. The content-based model represents category, genres, year bucket, director and actor tokens and ranks candidates by cosine similarity to a duration-weighted user profile. The association model mines programme pairs from training histories and scores candidates using co-view confidence and lift. Every method uses the same temporal validation protocol and excludes programmes already watched by the user; users without usable history use the popularity fallback.

3. Validation result

Method	MAP@5
popularity	0.162263
content	0.008688
association	0.157865

The final selected method is association, selected solely from the temporal validation MAP@5. The association score is close to the popularity reference, while the content model is weak on this sparse, metadata-heavy catalogue.

4. Two-user comparison

User 8377619604347126107: 90 labelled programmes in the joined history.

Method	Top-5 IDs and programme metadata
popularity	2500413 — Гражданская оборона. (Инфо) 700389 — Кто сверху?, 6 сезон, 7 эп. (Развлечения) 10002893 — Т/с 'Когда мы дома'. (Сериалы) 5800509 — Реальная мистика. (Развлечения) 500281 — Ток-шоу 'Говорит Украина'. (Развлечения)
content	90079323 — Х/ф 'Иллюзия обмана 2'. (Фильмы) 90077525 — Х/ф 'Пластик'. (Фильмы) 90072921 — Х/ф 'Большой солдат'. (Фильмы) 90078430 — Х/ф 'Kingsman: Секретная служба'. (Фильмы) 90079464 — Х/ф 'Дэдпул'. (Фильмы)
association	2500413 — Гражданская оборона. (Инфо) 700389 — Кто сверху?, 6 сезон, 7 эп. (Развлечения) 10003159 — Т/с 'Ничто не случается дважды', 1 и 2 с. (Сериалы) 700390 — Суперинтуиция, 3 сезон, 4 эп. Андрей Бедняков и Юлия Са (Развлечения) 500281 — Ток-шоу 'Говорит Украина'. (Развлечения)

User 8381667675275833309: 53 labelled programmes in the joined history.

Method	Top-5 IDs and programme metadata
popularity	2400508 — Сегодня. Итоги с Олегом Панютой. (Инфо) 20088 — 'Завтрак с 1+1'. Информационно-развлекательная программ (Развлечения) 200337 — 'Утро с 'Интером'. (Развлечения) 6200371 — 'ДжеДАИ'. (Познавательное) 400335 — 'Хата на тата', 7 сезон, 1 эп. (Развлечения)
content	500368 — Ток-шоу '60 минут' с Ольгой Скабеевой и Евгением Поповы (Развлечения) 90082757 — Х/ф 'Дельфин Берни'. (Фильмы) 90082758 — Х/ф 'Дельфин Берни 2'. (Фильмы) 90076672 — Х/ф 'Курьер из рая'. (Фильмы) 90081355 — Х/ф 'Невероятные приключения Факира'. (Фильмы)

association	2400508 — Сегодня. Итоги с Олегом Панютой. (Инфо) 20088 — 'Завтрак с 1+1'. Информационно-развлекательная программ (Развлечения) 200337 — 'Утро с 'Интером'. (Развлечения) 400335 — 'Хата на тата', 7 сезон, 1 эп. (Развлечения) 6200371 — 'ДжеДАИ'. (Познавательное)

5. Strategy and limitations

For this run I would deploy association scoring as a candidate generator with the popularity list as a fallback, then add a lightweight content re-ranker when metadata quality is sufficient. The association list stays close to popularity on validation, but uses the user history to alter ordering. The content model exposes the limitation of the catalogue: most schedule rows have no programme ID and metadata is uneven. The view log records platform behaviour rather than enjoyment, absence from the log does not prove a programme was never seen, and no A/B or revenue claim follows from this offline evaluation.

6. Reproducibility and submission

The implementation is in exam_sweettv/run.py. Generated artefacts include validation_results.json, audit.json, examples.json and submission.csv. The submission contains 1,260 rows and five space-separated IDs per row. Runtime environment: Python standard library, deterministic ranking and no random seed. Kaggle private-score screenshot and the original AI-session export must be attached before final submission.